Transfer of fixed review thresholds for breast ultrasound segmentation quality control

Running title Fixed review thresholds in breast ultrasound

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Abstract

Objectives To assess whether fixed numerical thresholds for segmentation review preserve detection and review workload after transfer between breast ultrasound datasets.

Methods A frozen Segment Anything Model used reference-derived, known-location boxes. Method A used predicted intersection over union; method B used disagreement after four box translations. Thresholds targeting 20 % review in 213 development cases were locked for 851 internal and 252 external cases. Low quality was reference Dice below 0.70. An external ranking analysis allocated 50 reviews per method.

Results Method A detected 71/93 (76.34 %) low-quality internal masks and 10/16 (62.50 %) external masks: −13.84 percentage points (95 % confidence interval, −38.85 to 7.45). Its review fraction changed from 174/851 (20.45 %) to 35/252 (13.89 %). At the locked thresholds, external detection was 10/16 for A and 13/16 for B; at 50 reviews, it was 14/16 and 13/16, respectively.

Conclusions Fixed-threshold transfer changed review workload, while external detection remained imprecise. The ordering of detection point estimates depended on whether numerical thresholds or review counts were held fixed. Evaluating both rules distinguishes operating-point transfer from ranking performance.

Keywords domain shift; external testing; prompt perturbation; selective prediction; vision foundation models

Introduction

The Segment Anything Model (SAM) can delineate an object from a spatial prompt, such as a box around a lesion [1]. Medical adaptations of SAM have evaluated box-prompted segmentation across several modalities, including breast ultrasound [2]. When a segmentation is used to support a measurement or other downstream analysis, a separate question is which outputs should be sent for review. A high overall Dice coefficient can coexist with a subset of masks that have poor agreement with a reference contour.

Case-level quality estimation without a new manual mask has been studied with reverse classification accuracy [3]. For promptable models, Kinalioglu used changes in mask structure under small prompt perturbations and combined them with model confidence for failure screening [4]. Loke et al. directly compared prompt-perturbation uncertainty with predicted IoU in SAM/MedSAM; most case-level differences remained unresolved after paired testing and multiplicity correction [5]. We examine a related operating question in breast ultrasound. If a service selects a risk-score threshold to review about one fifth of its development cases, the same numerical threshold may flag a different fraction and detect a different fraction of low-quality masks when applied to a second dataset.

We evaluate transfer of fixed numerical review thresholds as an operating policy for segmentation quality control, using model-confidence and prompt-consistency risk signals applied to identical masks from a frozen SAM. After calibrating each cutoff to a common 20 % development review target in BUS-BRA, we apply it unchanged to internal BUS-BRA and external BrEaST cases. Our contribution is to examine low-quality-mask detection, review fraction and missed cases jointly under this transfer, with external equal-review ranking as a distinct comparison. The prespecified primary contrast, H1, is the external-minus-internal difference in detection for method A. H2 examines its review fraction, and H3 compares B with A on the same low-quality external masks at the locked cutoffs. This design addresses the transfer of an operating point rather than introducing a new segmentation or uncertainty algorithm.

Materials and methods

Study design and datasets

This retrospective evaluation used anonymized images and manual lesion contours from two public breast ultrasound datasets. Reporting was informed by the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) 2024 update [6]. The patient was the analysis unit, with one image and one primary reference mask per patient. Figure 1 shows the development, internal test and external test cohorts. Table 1 describes their composition.

BUS-BRA version 1.0 contains 1,875 images from 1,064 patients [7, 8]. We used a patient-level split stratified by benign or malignant lesion status to obtain 213 development and 851 internal test patients (seed 20261105). We selected one eligible image per patient using seed 20261106. The two cohorts had disjoint patient identifiers. We verified image and mask pairs and used the actual pixel dimensions when these differed from metadata. An initial 40-case feasibility run used a subset of development patients. The full development set supplied the formal thresholds.

The BrEaST version 1 archive includes 256 scans, each from a different patient [9, 10]. We matched the official image and main tumor mask by case identifier. Four normal cases lacked a primary lesion mask, leaving 252 external test patients. The primary reference was the provided main tumor contour. Fourteen additional masks were outside that reference definition. The audit found zero exact image matches between the datasets.

[Place Table 1 near here]

Prompted segmentation and risk scores

We used the official SAM ViT-B checkpoint (sam_vit_b_01ec64.pth) with frozen float32 weights and single-mask output [1]. The original source commit, checkpoint hash and software versions are recorded in Table S1. We decoded images to red, green and blue (RGB) channels. BrEaST images were alpha-composited over black before conversion to RGB. This changed the RGB pixel values of eight images relative to discarding the alpha channel. Reference masks were read as binary arrays.

We obtained the initial prompt box from the tight bounding box of the nonempty reference mask, expanded each horizontal side by 10 % of the width and each vertical side by 10 % of the height, and clipped the result to the image. This is a reference-informed, known-location (oracle-localization) setting, not a test of autonomous lesion localization. We encoded the image once and used the resulting embedding for the initial prediction and four additional predictions (Figure 2). The fixed 10 % context margin was used consistently across datasets.

Method A calculated risk as one minus the raw predicted intersection over union (IoU) returned for the initial mask. For method B, we shifted the initial clipped box left and right by 5 % of its width and up and down by 5 % of its height, clipping each shifted box to the image. We calculated the Dice coefficient between the initial predicted mask and each of the four shifted-prompt masks. Risk B was one minus the mean of these four Dice coefficients. We retained all four directions when clipping produced coincident boxes. Two empty predicted masks had a consistency Dice coefficient of 1. The 5 % translations define a deterministic local stability probe relative to box size. Method B adds four mask-decoder passes while reusing the image embedding. Recorded inference times cover the combined pipeline; A-only and incremental B times were not separately benchmarked.

For binary masks, the Dice coefficient is twice the number of shared foreground pixels divided by the sum of foreground pixels in the two masks. The quality reference for both methods was Dice between the initial prediction and the manual mask. An initial empty prediction against a nonempty reference contributed a Dice coefficient of 0. We defined low quality as reference Dice below 0.70, an operational cutoff for this evaluation.

OpenAI Codex assisted with protocol discussions and implementation of research, statistical and plotting code. The scoring pipeline processed the public images using frozen SAM weights, and figures were rendered conventionally from recorded data. Both authors reviewed the implementation and calculated outputs. A precise underlying Codex model identifier was not available in the archived study records.

Development threshold selection

For each method, we searched the distinct development risk scores and the option to flag zero cases. A case was flagged if its score met or exceeded the candidate threshold. The selected threshold gave the

review fraction closest to 20 %. Cases tied at the cutoff were flagged together. When two choices were equally close to the target, we selected the one with fewer flagged cases. The exact thresholds were locked before test scoring and applied to both test cohorts. The 20 % target defines a prespecified one-in-five review scenario rather than an empirically measured clinical review capacity.

Outcomes and statistical analysis

The analysis included cases with a valid reference Dice and both finite risk scores. The procedure allowed one retry with the same configuration after a technical failure and recorded persistent failures separately. We report low-quality detection as the number of flagged low-quality masks divided by the number of low-quality masks. Simulated review is the number of flagged cases divided by all scored cases. Misses per 100 patients is 100 times the number of unflagged low-quality masks divided by all scored cases.

The dated analysis plan specified three contrasts. H1, the primary contrast, compared method A low-quality detection in BrEaST minus BUS-BRA internal testing. H2 used the same cohort order for method A simulated review. H3 assessed whether method B detected a larger fraction of the same low-quality BrEaST cases when the two cutoffs gave equal review counts in development. The paired contrast was B minus A. Differences are expressed in percentage points. We calculated two-sided 95 % confidence intervals (CIs) using Newcombe score method 10 for independent proportions in H1 and H2 [11] and its paired counterpart in H3 [12]. H2, H3 and supplementary intervals are nominal, with no multiplicity adjustment.

We repeated the detection analysis with reference Dice cutoffs of 0.50 and 0.80 while retaining both locked risk thresholds. We also ranked the external cases by each risk score and selected the top 50 per method, the count closest to a 20 % review fraction among 252 cases. We broke risk ties by lexicographic ordering of the official case identifiers. This equal-review comparison used a common number of reviews in the observed external cohort. Its paired interval is conditional on the observed rankings.

As post hoc descriptive quality-control measures, we additionally tabulated true-positive (TP), false-positive (FP), true-negative (TN) and false-negative (FN) counts, sensitivity (TP/(TP+FN)), specificity (TN/(TN+FP)), positive predictive value (PPV; TP/(TP+FP)) and negative predictive value (NPV; TN/(TN+FN)). Low-quality segmentation was the positive condition and a review flag the positive test. We used Wilson 95 % intervals for these proportions. These measures assess mask-quality screening, not diagnosis of breast cancer.

We additionally performed a post hoc exploratory stratification by dataset-reported benign/malignant classification, retaining the locked risk thresholds and Dice < 0.70 definition. BUS-BRA labels came from the official Pathology field, linked by image and patient identifiers. BrEaST labels came from Classification, linked by CaseID and verified against the selected filenames; 197 diagnoses were biopsy-confirmed and 55 were confirmed by follow-up. We report stratum counts, review fractions and detection with Wilson 95 % intervals. No stratum-specific recalibration, hypothesis test or interaction test was performed. We also audited all external low-quality cases missed by A, partitioned by the B review flag.

Computational reproducibility

We assessed computational reproducibility through two routes. We recomputed the statistical analyses directly from the archived case scores and reran the locked pipeline on all 1,316 study images in the same computing environment. We compared the new per-case Dice coefficients, predicted IoU, both risk scores, quality labels and review decisions with the original records. We also compared the recalculated calibration, statistical results and review-curve coordinates. Runtime was recorded separately.

Results

Cohort yield and development calibration

All 213 development, 851 internal test and 252 external test cases were scored. Low-quality masks occurred in 24/213 (11.27 %) development cases, 93/851 (10.93 %) internal cases and 16/252 (6.35 %) external cases. Each method flagged 43/213 (20.19 %) development cases. The locked thresholds were 0.13705134391784668 for A and 0.06972856653991777 for B.

No initial prediction was empty in the 1,316 study images. All 5,264 initial-to-shifted consistency Dice values were strictly positive, establishing that all shifted predictions were also nonempty. Thus the empty-empty Dice convention did not affect any recorded risk score or review decision.

Performance at the locked thresholds

For H1, method A flagged 71 of 93 low-quality masks in BUS-BRA internal testing (76.34 %) and 10 of 16 in BrEaST (62.50 %; Wilson 95 % CI 38.64 to 81.52 %). The external-minus-internal difference was −13.84 percentage points (95 % CI −38.85 to 7.45). The interval includes zero and is compatible with both a substantial decrease and a smaller increase in detection (Tables 2 and 3 and Figure 3).

[Place Tables 2 and 3 near here]

For H2, the method A simulated review fraction was 174/851 (20.45 %) internally and 35/252 (13.89 %) externally, a difference of −6.56 percentage points (nominal 95 % CI −11.25 to −1.10). Among masks with reference Dice of at least 0.70, A flagged 103/758 (13.59 %) internally and 25/236 (10.59 %) externally. The overall rates of missed low-quality masks with A were 2.59 per 100 internal patients and 2.38 per 100 external patients.

For H3, method B flagged 13/16 (81.25 %; Wilson 95 % CI 56.99 to 93.41 %) low-quality BrEaST masks compared with 10/16 (62.50 %) for A. Ten were flagged by both scores, three by B alone and three by neither. The paired difference was 18.75 percentage points (nominal 95 % CI −4.28 to 40.04). B flagged 33/252 (13.10 %) external cases and missed three low-quality masks. In the internal cohort, B flagged 150/851 (17.63 %) cases, detected 68/93 (73.12 %) low-quality masks and missed 25 (Table 2).

Sensitivity and equal-review comparisons

At reference Dice below 0.50, A detected 52/53 internal and 5/7 external low-quality masks, and B detected 43/53 and 5/7, respectively. At Dice below 0.80, the corresponding counts were 109/183 and 21/50 for A and 95/183 and 21/50 for B. With each alternative definition, A's external detection point estimate was lower than its internal estimate, and both methods had the same external detection count (Tables S2 and S3 and Figure S1).

At 50/252 reviews (19.84 %) per method in BrEaST, A detected 14/16 (87.50 %) and B detected 13/16 (81.25 %) low-quality masks. The paired B-minus-A difference was −6.25 percentage points (nominal 95 % CI −31.61 to 19.57). The order of the two detection point estimates differed between the locked-threshold and equal-review settings (Figure 4 and Table S4). Figure S2 shows the paired detections and misses in both settings.

Reproducibility checks

The raw-image rerun scored all 1,316 cases. Across the three cohorts, the maximum absolute difference was zero for each of the eight numerical fields compared. The quality and review decisions matched for every case (Table S5). Both routes reproduced the development thresholds, primary and supplementary statistics, and the count-level coordinates of the review curves.

Additional descriptive quality-control measures

The post hoc external quality-control counts were TP=10, FP=25, TN=211 and FN=6 for A, and TP=13, FP=20, TN=216 and FN=3 for B. Specificity was 89.41 % for A and 91.53 % for B; PPV was 28.57 % and 39.39 %, and NPV was 97.24 % and 98.63 %, respectively. The complete counts and Wilson intervals are provided in Tables S6 and S7.

Exploratory diagnostic strata

Internal benign- and malignant-labelled groups contained 577 and 274 patients, including 53 and 40 low-quality masks. A detected 51/53 (96.23 %; Wilson 95 % CI 87.25 to 98.96 %) and 20/40 (50.00 %; 35.20 to 64.80 %), while B detected 41/53 (77.36 %) and 27/40 (67.50 %), respectively. Externally, each diagnostic stratum contained eight low-quality masks: A detected 5/8 in each group, and B detected 7/8 benign-labelled and 6/8 malignant-labelled cases. Review fractions and intervals are reported in Table S8. All stratum counts recombine exactly to the overall results.

External missed-case audit

The complete set of six external low-quality cases missed by A comprised three also missed by B (cases 089, 175 and 238) and three detected by B (024, 105 and 222). In the shared-miss group, mean prompt-consistency Dice was 0.9590, 0.9799 and 0.9942, respectively, despite reference Dice of 0.3191, 0.0293 and 0.6813. Table S9 reports all six cases, their archived scores and flags. This is an outcome-defined descriptive case audit, not an additional estimate of population performance.

Discussion

Transfer of the review threshold

The method A threshold selected near the 20 % development review target flagged approximately one fifth of the internal cohort and approximately one seventh of BrEaST. The primary estimate for external detection was lower, but the confidence interval was wide with only 16 low-quality external masks. These results give different levels of precision for the two operational questions. The simulated review fraction was estimated from all 252 external patients, while detection was estimated from the 16 whose masks fell below the quality cutoff.

The review fraction depends on the mix of low-quality and acceptable masks and on how often each group is flagged. BrEaST had both a smaller low-quality fraction and a smaller flagged fraction among

acceptable masks for A. Misses per 100 patients similarly depend on the frequency of low-quality masks. This explains how the external point estimate for A detection could decrease while the number of misses per 100 patients remained close to the internal estimate. The dataset comparison combines differences in case mix, image acquisition and reference annotations, and cannot identify their separate effects.

The exploratory strata make the case-mix question more concrete: internal confidence-based screening captured different fractions of low-quality masks in the benign- and malignant-labelled groups. The external A point estimates were equal across labels, with eight low-quality events in each group. These descriptive associations do not establish a diagnostic-class mechanism for the cohort difference; they identify strata for prespecified evaluation in a larger independent sample.

What prompt disagreement adds

In the locked external evaluation, the disagreement score flagged three low-quality masks missed by the confidence score. The paired interval leaves the size and direction of the underlying difference uncertain. At the same external review volume, the point estimate favored A, and the two alternative quality cutoffs yielded equal external detection counts. The observed advantage of B at its locked threshold therefore depends on the evaluation rule and review volume.

The two scores capture different properties of the same mask. A uses the quality estimate returned with the initial prediction, while B measures stability to a specific set of box translations. A mask can be stable under those translations and still differ from the manual contour. Prior prompt-response work combined structural changes with confidence and reported ranking-based screening [4]. Direct comparisons of perturbation uncertainty and predicted IoU also predate this study [5]. Here, keeping the two scores separate and freezing the development cutoffs allows detection, review volume and misses to be inspected together on the second dataset.

Limitations and implications

The reference contour supplied the localization box and the target for quality assessment. The findings apply to this known-location setup and may differ for reader-drawn or automatically generated boxes. Only one SAM checkpoint, one image per patient and one external dataset were evaluated. Image-level overlap between the evaluation archives and the model's SA-1B pretraining corpus could not be determined from public records [1]. The small number of external low-quality masks limits precision, particularly for the paired comparison.

The Dice cutoff defines agreement with the released contours. A clinical standard for acceptable boundary error would require separate justification. Simulated review is a count of flags. Reader time, the success of mask correction and effects on downstream measurement would require evaluation with readers. Future testing should estimate workload and low-quality detection jointly under independently supplied prompts and across additional datasets.

Although the empty-empty convention was not invoked in this dataset, a future case with empty initial and shifted predictions could receive zero disagreement risk despite zero reference Dice. An explicit empty-mask review rule would be a prospective change; it was not added to the locked analysis.

In this known-location breast ultrasound evaluation, carrying forward fixed numerical cutoffs changed the fraction of cases flagged for review. External detection was estimated from 16 low-quality masks and

remained imprecise. The higher detection point estimate for B at the locked cutoffs did not persist when both scores received 50 external reviews. Reporting detection, review fraction and misses jointly, and specifying whether a cutoff or a review count is held fixed, distinguishes operating-point transfer from ranking performance. This distinction provides the practical interpretation of the present results.

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AI use and acknowledgments. OpenAI Codex assisted with study-design discussions, research and statistical code, plotting scripts, and substantive drafting and editing. Figures were rendered with Python/Matplotlib from released images, saved masks and recorded scores. Editorial drafting used the Scientific Agent Skills workflow [13]. A precise underlying Codex model identifier was not available in the archived study records. Both authors reviewed the research code, data, calculations, figures and text and accept responsibility for the work.

Research ethics. This retrospective secondary analysis used lawfully obtained, publicly released anonymized breast ultrasound images and annotations and involved no participant recruitment or intervention. The authors confirmed that the study met the conditions for exemption from ethics review under Article 32 of the Chinese Measures for Ethical Review of Life Science and Medical Research Involving Human Participants (2023).

Author contributions. All authors have accepted responsibility for the entire content of this manuscript and approved its submission. Haoze Lan: research conceptualization and principal study design. Ziyuan Zhu: code review, GitHub repository uploads and manuscript revision. Both authors completed their review of the code, data, figures and text.

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Conflict of interest. The authors state no conflict of interest.

Informed consent. No participants were recruited or contacted for this secondary analysis of publicly available anonymized data, and no additional informed consent was obtained.

Data and code availability. The original BUS-BRA version 1.0 images and reference masks are available at https://doi.org/10.5281/zenodo.8231412 and BrEaST version 1 at https://doi.org/10.7937/9WKK-Q141 [8, 10]. Study-specific case manifests, derived case-level scores, scoring and statistical scripts, and archived reproducibility records are publicly accessible in the research repository [14], archived study commit 592566b4c5f39e261c676eb034139a974e9cdfd5. The data, experiments and reproducibility directories distinguish original public datasets from newly derived study materials. No model checkpoint is bundled; its official download route and SHA-256 are documented in the repository. Additional exploratory stratum tables, missed-case records and executable audit scripts accompany the supplementary materials.

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Tables and table legends

Table 1. Study cohorts

Cohort	Patients, n	Benign, n	Malignant, n	Low-quality masks, n/N (%)
BUS-BRA development	213	145	68	24/213 (11.27 %)
BUS-BRA internal	851	577	274	93/851 (10.93 %)
BrEaST external	252	154	98	16/252 (6.35 %)

One image was selected per patient. Low-quality masks had a reference Dice coefficient below 0.70. n, number of patients; N, cohort size.

Table 2. Test performance at the locked review thresholds

Test cohort	Score	Detection, n/N (%) [Wilson 95 % CI, %]	Simulated review, n/N (%)	Misses, n	Misses per 100 patients
BUS-BRA internal	A	71/93 (76.34 %) [66.77, 83.83]	174/851 (20.45 %)	22	2.59
BUS-BRA internal	B	68/93 (73.12 %) [63.32, 81.08]	150/851 (17.63 %)	25	2.94
BrEaST external	A	10/16 (62.50 %) [38.64, 81.52]	35/252 (13.89 %)	6	2.38
BrEaST external	B	13/16 (81.25 %) [56.99, 93.41]	33/252 (13.10 %)	3	1.19

Detection uses low-quality masks as its denominator. Simulated review and misses per 100 use all scored patients in each cohort. A, model-confidence score; B, prompt-disagreement score; n, number of events; N, relevant denominator. Brackets show Wilson 95 % intervals for detection, in percent.

Table 3. Planned contrasts at the locked review thresholds

Analysis	Contrast	Difference, pp	95 % CI, pp
H1 (primary)	Method A detection, external minus internal	−13.84	[−38.85, +7.45]
H2 (secondary)	Method A simulated review, external minus internal	−6.56	[−11.25, −1.10]
H3 (secondary)	External detection, Method B minus Method A	+18.75	[−4.28, +40.04]

pp, percentage points. CI, confidence interval. H1 and H2 use independent Newcombe intervals. H3 uses a paired interval in the 16 low-quality BrEaST cases. H2 and H3 intervals are nominal.

Figure legends

Figure 1. Study cohorts and analysis flow. BUS-BRA patients were assigned to development or internal testing. BrEaST contributed external test patients with a primary lesion mask. Each patient contributed one image. Low-quality masks had a reference Dice coefficient below 0.70.

Figure 2. Review scores for prompted breast ultrasound segmentation. (A) The initial box is derived from the manual contour with a 10 % margin per side and clipped to the image. Four further prompts shift the clipped box by 5 % of its width or height. (B) Frozen Segment Anything Model (SAM) ViT-B reuses one image embedding to generate one mask per prompt. (C) Method A uses one minus predicted intersection over union for the initial mask. Method B uses one minus the mean Dice between the initial and four shifted masks. Each score triggers simulated review at or above its locked cutoff. The image and masks share a display crop from BrEaST case 001 [10].

Figure 3. Planned differences at the locked thresholds. Points show differences in proportions and horizontal bars show two-sided 95 % Newcombe confidence intervals. H1 and H2 compare BrEaST minus BUS-BRA internal testing. H3 compares B minus A in the same 16 low-quality BrEaST cases. H2 and H3 intervals are nominal. The dashed line marks zero difference.

Figure 4. Low-quality mask detection by simulated review volume. Panels A and B show the percentage of low-quality masks detected as progressively more patients are selected in descending risk order in each test cohort. Markers identify the locked thresholds and the equal-review setting. Panel C compares detection in the same 16 low-quality external cases at the locked thresholds and at 50 reviews per method.